



An integrated framework for effective service and repair in the automotive domain: An application of association mining and case-based-reasoning

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ABSTRACT

In this paper, we propose a novel integrated framework combining association rule mining, case-based-reasoning and text mining that can be used to continuously improve service and repair in an automotive domain. The developed framework enables identification of anomalies in the field that cause customer dissatisfaction and performs root cause investigation of the anomalies. It also facilitates identification of the best practices in the field and learning from these best practices to achieve lean and effective service. Association rule mining is used for the anomaly detection and the root cause investigation, while case-based-reasoning in conjunction with text mining is used to learn from the best practices. The integrated system is implemented in a web based distributed architecture and has been tested on real life data.

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1. Introduction

In today's global market there is a continuous pressure on the Original Equipment Manufacturers (OEM's) to improve service processes that directly relate to the customer satisfaction. After-sales -service is a critical business process that has a significant impact on the consumer's perception about the brand and the product quality. A recent study by J.D. Power and associates revealed that customer satisfaction with the dealer service and the cost of service/repair have significant impact (over 20%) on the notion of customer satisfaction [1]. Also, incorrect service results in poor QRD (quality, reliability and durability) that in turn increases customer dissatisfaction. On the other hand, in the last couple of decades, the use of electronic systems in automobiles has grown rapidly. This growth is driven by the growing requirements for environmental protection and demands from the consumers to improve fuel economy, comfort and safety. As a result, the amount of software embedded in a vehicle increased to tens of millions of lines of codes [2]. Such a complex nature of modern vehicle systems often restricts service technician's ability to diagnose the problem accurately in a short time. This problem further

aggravates for new/inexperienced technicians. To maintain competitive edge in today's market, it is essential to quickly identify technician difficulties, identify their training needs, and provide appropriate training to diagnose and repair the faults in a cost-effective way. This paper presents a novel integrated framework that achieves these objectives for continuous improvement facilitating lean service in the automotive domain.

The notion of 'lean' paradigm has been well established and practiced in the manufacturing domain for the last few decades. It involves: (a) understanding customer value, (b) identifying the value stream that adds to this value, (c) creating flow of product and information, and (d) seeking a wasteless perfection [3]. Brunt and Kiff [4], Kiff [5] described how these principles can be adapted in the service industry for improving the performance related to dealer's service such as delays, defects, rework, unnecessary movements, and excess stock, among others. In the current work, we focus on improving the factors related to customer satisfaction, in particular, *improvement in mean time to repair, first time fix, and avoidance of unnecessary repairs*. To this end, we make use of association rule mining to detect the anomalies (possible wastes) in the field and identify their root causes. Then, we subscribe to the case-based-reasoning (CBR) and text mining to retrieve correctly solved cases (i.e. best practices) from the past and learn from them, and further transfer the learnt (discovered) knowledge to technicians to achieve continuous improvement in the service and repair.

In Section 2, we provide state-of-the-art review of different disciplines that are used in our approach. Then, in Section 3, we

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describe the architecture of our framework in detail. It involves (a) identification of different types of anomalies related to fault diagnosis and repair, (b) application of association rule mining to correctly detect the root causes associated with the anomalies, and (c) intelligent learning and a case-based-reasoner to learn the best-practice knowledge from the correctly solved cases. In Section 4, we provide results and discuss the performance of our system. Finally, in Section 5, we conclude this paper by highlighting the main contributions of this work.

2. Related work

As described earlier, to achieve our objective of lean service, we propose a novel hybrid system that makes use of the principles of data mining and knowledge discovery – especially, association rule mining, text mining, and case-based-reasoning. Related work in these different streams of technologies is discussed here.

Agarwal et al. [6] formalized the notion of association rule mining to find the associations between two sets of products in a transaction database using Artificial Immune Systems (AIS) and Apriori algorithms. In their approach, if 'I' represent the set of product items and 'T' represent the set of transactions, where each transaction includes a set of products that are purchased together, then the association rule mining establishes the associations of the form: $X \rightarrow Y$, where $X \in I$ and $Y \in I$. Several association measures such as support, confidence, conviction, lift, coverage, correlation, and odds ratio have been reported in the literature [7]. However, *support* and *confidence* are the most popularly used [8]. The *support* is the percentage of transactions that contains both X and Y, whereas, the *confidence* is the fraction of transactions containing X that also contains Y. A notion of Apriori algorithm has been proposed to minimize the computational complexity, especially, while dealing with very large databases. The Apriori algorithm involves finding frequent item sets (with previously assigned level of support) and then generating the rules that comply with previously assigned level of confidence. Recently, Buddhakulsomsiri et al. [9] developed association rule mining algorithms using Apriori and elementary set concept techniques to find the relationships between the product attributes (such as production date, repair date, mileage, engine type) and the causes of failure (related labor code) for warranty data analysis.

The case-based-reasoning is an analogical reasoning method which relates previously solved problems or experiences to the current unsolved problem. Generally stating, the CBR approach consists of the following steps – (a) storing successfully solved cases in the form of problems and solutions in a case-base, (b) retrieving similar past cases based on a set of attributes of a problem at hand and using the solution of past cases to solve new problem, and (c) adapting (if necessary) the solution of a previous case to the new problem [10,11]. The CBR methodology became popular in the last few years mainly because of the following benefits: (a) it enables finding solutions to complex problems quickly, (b) captures the experience of a skilled specialist, and, (c) provides a basis to discover the decision knowledge from the hidden data [12]. The success of a CBR system depends on its ability to retrieve the most relevant case in support of a solution to a new problem. A number of methods have been proposed for the correct case retrieval, and some influential ones include induction, knowledge-based indexing, and the nearest neighbor algorithms [13]. Further, Textual CBR (TCBR) [14] and text mining techniques have been developed and applied in situations where experiences are predominantly captured in the form of text. Typically, the text mining tasks consists of text categorization, text clustering, concept/entity extraction, relationship extraction between concepts, production of taxonomies, and document summarization.

Recently, a significant amount of work has been reported on application of the case-based-reasoning and text retrieval techniques for fault diagnosis and troubleshooting to improve maintenance and service. Park et al. [15] developed a decision support system using the CBR approach to accumulate knowledge for solving problems in a complex production system. Yang et al. [16] integrated the CBR with an Adaptive Resonance Theory-Kohonen Neural Network (ART-KNN) to enhance fault diagnosis. In this case, the Neural Network is used to guide the CBR module to search for the similar previous cases. Saxena et al. [17] used the CBR approach for integrated fault diagnosis and health maintenance architecture for fleet vehicles. Liu and Ke [18] developed a knowledge support framework for solving problems in production environment using the CBR and text mining. Chougule and Chakrabarty [19] employed the CBR using semantically enhanced ontology guided search to retrieve fault diagnosis knowledge in a vehicle assembly plant. Wang and Hsu [20] integrated Cognitive Task Analysis (CTA), hierarchical clustering and ontologies for PC troubleshooting knowledge management systems. Jin et al. [21] used domain specific ontology to represent facility fault diagnosis knowledge and used it for fault diagnosis.

In summary, the association rule mining with the Apriori algorithm has been used for automotive warranty data analytics. However, one of the limitations of such approach is that infrequent associations are likely to be overlooked. To alleviate this problem constraint based association rule mining has been used in the current work. The incorporation of the domain specific knowledge (constraints) helps to reduce the computational complexity as well as noise while identifying the meaningful associations between data. Such incorporation of the domain specific knowledge is missing in previous work. The CBR and text mining have become popular for fault diagnosis and troubleshooting over the last few years. Several applications of the CBR and ontology have been reported for machine (manufacturing) fault diagnosis. However, it appears that very limited work has been reported on application of the CBR to automotive domain which poses its own challenges. In particular, automotive domain provides an acid test for the CBR technique in retrieving the correct k-nearest cases for solutions that are associated with millions of vehicles in the field. The search space becomes more complex because there are thousands of technicians involved in the service and repair of millions of vehicles. Firstly, it is essential to identify proper cases involving accurate fixes in a cost effective way. Secondly, often a significantly large number of similar cases are retrieved from the archived case-base and it is essential to aggregate the knowledge derived from these retrieved cases to suggest possible solutions. As a result, in the current work, we make use of the inductive and knowledge guided approach to correctly retrieve the past cases and learn from them. In addition, the text mining approach is used for solution adaptation and learning. It appears that, a seamless integration of the association rule mining, case-based-reasoning and text mining techniques for maintenance and service improvement is not reported earlier.

In the next section, we describe the overall architecture of our proposed framework that combines the association rule mining, case-based-reasoning and text mining for the anomaly detection, its root cause investigation and learning from the best practices in the field to achieve continuous improvement in the service and repair.

3. Framework architecture

When an abnormal behavior (or fault) is detected in a vehicle, diagnostic trouble codes (DTCs) are generated and stored on the vehicle memory bus. At dealers' end, a technician extracts these DTCs to identify the root cause of anomalies and determine the

repair actions that need to be performed on that vehicle. Usually, the predefined codes, i.e. labor codes (LC), which characterize the nature of repair actions are assigned to the repairing tasks that are performed by a technician to fix a vehicle problem. In addition to this, the information related to labor time, parts changed, dealer (technician) involved and technician verbatim entries are also maintained as useful data. The technician verbatim describes the textual summary of the actual on-field repair tasks performed by a technician. It typically consists of the symptoms that are observed with a specific part and the repair actions that are taken to fix the root cause of symptoms. Fig. 1 shows overall architecture of our framework at a coarse-grained level, which consists of following components – *anomaly detector*, *case classifier*, *root cause analyzer* and, *intelligent learning and reasoner*.

The *anomaly detector* takes as input the data coming from the two databases (Symptoms and Claims) and detects different types of anomalies, such as, cases with high time to repair or the cases with repeat visits. The Symptoms database contains data related to vehicle, DTC and the date on which a particular DTC is read. The Claims database contains data related to vehicle, labor codes (LCs), the date on which specific repairs are performed along with technician verbatim that are written in a free flowing natural English language. The *anomaly detector* makes use of the Claims database to identify following types of anomalies – (a) repair cases in which a technician has taken high time to repair, (b) repeat visit cases where a vehicle visits a dealer again within a short period of time, and (c) cases involving high cost repair, e.g. unnecessary part changes. Having identified such anomalies, the remaining cases that are residing in the database represents cases in which the technicians have correctly solved the problem in a cost-effective way. The *case classifier* takes as an input the anomalous cases detected by the *anomaly detector*, and also the correctly solved cases, and classifies them into two buckets. The first bucket

represents instances of correctly analyzed cases, that is, the cases in which a vehicle related problems are correctly fixed by technicians. The second bucket represents instances of incorrectly analyzed cases by technicians at different dealers. The successfully solved cases (first bucket) form the “case-base” in the learning and the reasoning process. The main aim of the *root cause analyzer* is to identify possible root causes, i.e. symptoms that are associated with the anomalous cases. And for this, it makes use of the association rule mining and text mining techniques. The symptoms identified by the *root cause analyzer* are then presented to the end user (the engineers and the domain experts from quality, product engineering and service). The *intelligent learning and reasoner* makes use of the case-based-reasoning technique, which takes these symptoms as an input and then retrieves similar cases from the archived case-base and then learns how such cases have been diagnosed correctly in cost-effective way. The text mining component facilitates solution adaptation and learning in cases where the inference from the retrieved cases cannot directly be applied to the problem at hand. To deal with this, the text mining engine analyzes the text written in the technician verbatim entries, especially, in order to find additional not-DTC symptoms recorded in the technician verbatim. This process of adaption and learning is performed interactively by the end user. Finally, the suggested solutions are presented to domain experts for validation. The validated knowledge is retained and disseminated to technicians to improve their performance in solving similar cases in a cost-effective way. In summary, main steps involved in process are outlined below and described in consequent subsections:

- Detection of anomalies in the field.
- Root cause analysis of anomalies using association rule mining.
- Reasoning and intelligent learning from non anomalous cases to identify best practices.

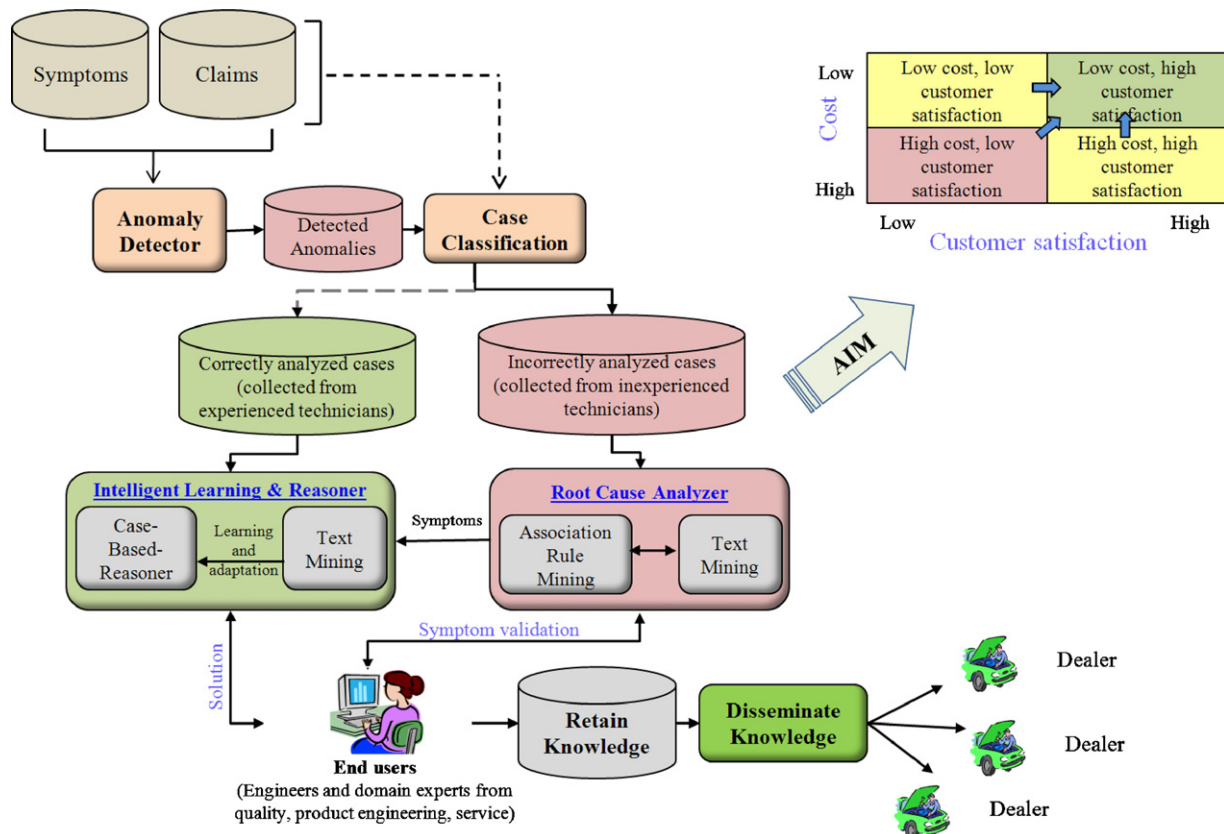


Fig. 1. The overall system architecture.

- Retain and disseminate newly discovered knowledge for continuous improvement.

The following sections describe each of these steps in detail.

3.1. Anomaly detection from on-field data

First step towards lean service is to successfully identify wastes and other adverse factors that are contributing to customer dissatisfaction, such as repairs contributing to high repair time, repairs contributing to repeat visits, and expensive repairs. The following section describes these anomalies and their detection.

3.1.1. Repairs contributing to high repair time

As mentioned earlier, the repair tasks that are performed by a technician are assigned to predefined labor codes (LCs) along with the repair time. Here, the anomaly is characterized in terms of the repair cases in which the time assigned to the labor codes is higher than the predefined standard time that has been determined by performing time study. Severity of labor codes pertaining to such anomalies is expressed in terms of the notion of *contribution*. It is defined as the ratio of summation of extra repair time taken for all the instances of a given labor code (say, L_i) to summation of extra repair time taken for all labor codes (including L_i). It is calculated by using the formula described below:

Let $L = \{L_1, L_2, \dots, L_n\}$ be the set of n distinct labor codes. Let t_i be the standard repair time for labor code L_i , where, $1 \leq i \leq n$. Let L_i occurs for m_i instances. Let t_{ij} be the repair time duration for j th occurrence of labor code L_i

Percentage contribution of labor code L_i in high repair time

$$= \frac{100 \times \sum_{k=1}^{m_i} (t_{ik} - t_i)}{\sum_{i=1}^n \sum_{k=1}^{m_i} (t_{ik} - t_i)} \quad (1)$$

The anomaly detection related to high repair time and the concept of contribution has been explained with a case study in Section 4.1.

3.1.2. Repairs contributing to repeat visits

The repeat visit is one of the important factors that contribute to customer inconvenience, especially, when a customer visits the dealer again within a short period of time with the same problem. Moreover, the repeat visits also play a crucial role in determining a perceived quality and reliability of the product. To this end, our aim is to minimize the number of repeat visits of customers by identifying the anomalies related to the repeat visits. In our approach, such anomalies are identified by using association rule mining approach. Here, we first identify the associations between the labor codes that are used during two consecutive visits of the same vehicle to the dealer.

Let $S1 = \{L_1^1, L_2^1, L_3^1, \dots, L_p^1\}$ be the set of labor codes (repairs) used during the first visit of vehicle V1.

Let $S2 = \{L_1^2, L_2^2, L_3^2, \dots, L_q^2\}$ be the set of labor codes (repairs) used during the second visit of vehicle V1.

We take the Cartesian product of the two sets, S1 and S2 to obtain possible associations between the repairs that are performed during the first and the second visit of vehicle V1.

Set of possible associations, $C = \{L_1^1, L_1^2\}, \{L_1^1, L_2^2\}, \dots, \{L_p^1, L_q^2\}$ (2)

Aggregation of such associations for all the vehicles within a specific period of time allows us to highlight major repairs¹ that are contributing to repeat visits to dealers.

¹ Recall here that each labor code represents a specific type of repair action performed by a technician on a specific part when a specific type of symptom is observed.

Table 1

Sample data for repeat visit anomaly detection.

Vehicle	Labor codes the first visit (S1)	Labor codes during the second visit (S2)	Associations between labor codes at vehicle level
V1	L1,L3	L2	{L1,L2},{L3,L2}
V2	L1	L2	{L1,L2}
V3	L3,L5	L5	{L3,L5},{L5,L5}
V4	L1	L2	{L1,L2}

Here, we show with an example how this approach is used to identify the repeat visit issues. Table 1 shows the repair data corresponding to two consecutive visits of 4 vehicles and the associations between the labor codes at each vehicle level. Aggregation of these associations shows that the repairs {L1,L2} are involved in maximum number of repeat visit cases (here, 3). It implies that, the technicians are not able diagnose for L1 properly during the first visit resulting into the second visit of the vehicle.

In the association rule mining, domain specific filters have been used to restrict analysis to a specific system or subsystem of a vehicle. The association results are also useful to extract the causal relations between the failures. These causal relations can be used to anticipate the future failures and take corrective action in the first visit itself. For example, it was observed that the vehicles that were repaired for 'power steering hydraulic hose replacement' were visiting to the dealer again within short period of time to perform the repair related to 'power steering pump assembly.' It is likely that, the leakage from the pipes can lead to malfunctioning of the pump assembly. Under such circumstances, dealers can be informed to check the pump assembly after replacing the hose. Such kind analysis and knowledge transfer helps to reduce the number of visits to the dealer.

3.1.3. Repairs with expensive repair cost

In addition to the repairs contributing to high repair time and repeat visits, another factor that causes inconvenience to the customers is unnecessary expensive repairs, especially, when the same problem can be solved by other cost-effective ways. For example, in some cases, a vehicle visits a dealer with a fault say 'loose mirror assembly' and an experienced technician fixes the fault by simply *repairing* the assembly. In some other cases, the same fault may get fixed through the *replacement* of a faulty assembly by inexperienced technicians. In such cases, it is crucial to learn the service knowledge from the repairs where technicians successfully fixed the faults by making optimal utilization of resources, i.e. cost and time.

3.2. Root cause analysis of anomalies using association rule mining

The main aim of the root cause analysis is to identify the frequent symptoms related to anomalous cases. The association rule mining technique has been used to find the associations between the data from the Symptoms (DTCs) and the Claims databases for the anomalous cases. Moreover, it also identifies whether any specific technicians (or dealers) are involved in the anomalous cases so that they can be trained to prevent occurrence of such anomalies in the future.

Before performing the association rule mining, it is necessary to select a correct set of data from the Symptom and the Claims databases and then filter the retrieved data such that it can be used for performing the association rule mining. The first step involved in association rule mining is to make use of the filter that is developed to select the data associated with a specific vehicle make, model, and year. It also takes into account the time difference between DTC read date and repair date such that the

association between only related DTCs and LCs are established correctly. Below we show how the associations are established between DTCs and LCs.

Let,

$V = \{V_1, V_2, \dots, V_r\}$ be the cases (vehicles) corresponding to anomaly.

$V_i = \{V_{i1}, V_{i2}, V_{i3}, \dots, V_{is}\}$, where, $V_i \subset V$, is a subset of vehicles with the anomalous labor code L_i .

$V_i^j = \{V_i^{j1}, V_i^{j2}, V_i^{j3}, \dots, V_i^{je}\}$, where, $V_i^j \subset V_i$, is a subset of vehicles with the symptoms S_i .

$S_i = \{DTC_{1i}, DTC_{2i}, \dots, DTC_{ui}\}$, where, DTC_{1i} to DTC_{ui} are the diagnostic trouble codes in S_i . In other words, S_i itself consists of multiple DTCs associated with it. As a result, it is a non-trivial effort to determine which labor codes can be used to fix the given set of symptoms to avoid high time to repair and repeat visits.

We perform the association rule mining to identify whether any specific symptoms S_i is a root cause of anomalous repair L_i .

P = Percentage of instances symptom S_i is root cause of the anomaly L_i

$$P = \frac{|V_i^j|}{|V_i|} \quad (3)$$

Here, we show with an example how this method is used to investigate the root cause of the anomaly. Table 2 shows one such set of data, which is used to perform the root cause analysis associated with a specific type of anomaly, say high time to repair. In this table, the 'Symptoms' column indicates the string of DTCs that are set for the anomaly given in the column 'Labor Code'. As it can be seen in the column 'Labor Code', the same labor code $L1$, appeared in cases {1, 2, 3, 4}. Similarly, as shown in column 'Symptoms' the same set of symptoms $S1$ consisting of DTCs {DTC1, DTC2, DTC3} appeared in cases {1, 3, 4}. As a result, in 75% of cases in the field, root cause of anomaly related to $L1$ is $S1$. Typically, in multiple DTCs scenario (i.e. with symptoms DTC1, DTC2, DTC3), an experienced technician takes less time to diagnose the problem by identifying the fact that there may be only a *single failure* that is causing all DTCs i.e. DTC1, DTC2 and

The CBR problem-solver consists of following main steps: (a) retrieve the similar cases to provide the most likely solution to solve the problem (symptoms), (b) validation of suggested solutions, and adaptation and learning if the solution cannot be used directly to solve the problem at hand, (c) retain and disseminate the solution (knowledge) for future use. Fig. 2 shows different steps that are used in the case-based-reasoning component of our system. These are described in detail in the following sub-sections.

3.3.1. Case retrieval

Retrieval of the correct case from the case-base is one of the most critical steps in the CBR process. In our approach, all the cases residing in a case-base consists of:

- Problem description: It describes the state of the world, that is, observed symptoms (DTCs), DTC read date, corresponding vehicle identification number (VIN), vehicle model, model year, vehicle engine type, transmission type and sale region. The rationale to choose these attributes is, DTCs as well as the conditions to set the DTC varies with the vehicle model (such as, Impala, Cadillac STS), model year (such as, 2009, 2010), engine type (example, 1.4 Liter Engine, 1.6 Liter Engine), transmission type (example, Automatic, Manual) and sale region (example, USA, Canada, China).
- Solution: The solution includes labor codes that are used to solve the problem in the past for each VIN, repair date and the technician verbatim corresponding to each labor code.

The case retrieval component of our system takes as an input the attributes related to the problem description such as product details and symptoms. These symptoms represent the DTCs that are difficult to diagnose and lead to anomalies in the field. The case retrieval component retrieves similar cases from the bucket containing correctly solved cases (cf. Section 3). Here, the cases with the exact match of these attributes are retrieved such that the value of similarity is either 1 or 0.

$$\text{Similarity}_{\text{input problem} \leftrightarrow \text{past case}} = \begin{cases} 1 & \text{(if all attributes related to product and DTCs match)} \\ 0 & \text{(if there is a mis-match in any one of the attributes)} \end{cases} \quad (4)$$

DTC3 in a vehicle system or subsystem. As a result, less diagnosis and repair time is needed to fix the single failure which in turn fixes all the DTCs. Comparatively, inexperienced technicians may take more diagnosis and repair time as they may tend to diagnose the faults for individual DTC in $S1$. As the claims data also contain information related to dealer involved in the repair, investigation can be done to identify whether any dealer (or technician) is involved frequently in the anomalous cases. This helps to identify the training needs at an individual technician level. The text mining of technician verbatim corresponding to these cases facilitates further investigation into the facts. The details of the text mining component are described in Section 3.3.2. In the following section, we describe the CBR component of the architecture for the intelligent learning from the best practices.

3.3. Reasoning and learning

The main aim of the reasoning and learning component of the system is to identify the correct procedures to fix the symptoms that are causing different anomalies and transfer the learnt knowledge to inexperienced technicians to improve the service. We subscribe to case-based-reasoning as a problem-solving technique that allows us to make use of the correctly solved cases in the past to improve the performance.

As it can be imagined, at any given instance there is a large population of vehicles present in the field. As a result, the case-base consists of several thousands of cases which potentially can be used to identify the solution for the symptoms and also there are multiple solutions that can be used to fix the symptoms in the field. Based on our experience we realize that, it is not always possible to retrieve the nearest neighbor solution by using the k-nearest neighbor algorithm because too many similar cases with varied solutions may get retrieved from population of millions of vehicles in the field. Under such circumstances, it is essential to aggregate the knowledge from retrieved cases to determine the likeliest solution. As a result, knowledge guided inductive reasoning and learning has been used to find the likeliest solution to the problem at hand. In our approach, we formalized the domain specific

Table 2
Sample data for root cause analysis of anomalies.

Cases (vehicles)	Labor code	Symptoms	Dealer/Technician
1	L1	DTC1, DTC2, DTC3	Dealer 1
2	L1	DTC6	Dealer 2
3	L1	DTC1, DTC2, DTC3	Dealer 1
4	L1	DTC1, DTC2, DTC3	Dealer 1
5	L2	DTC7, DTC8	Dealer 2

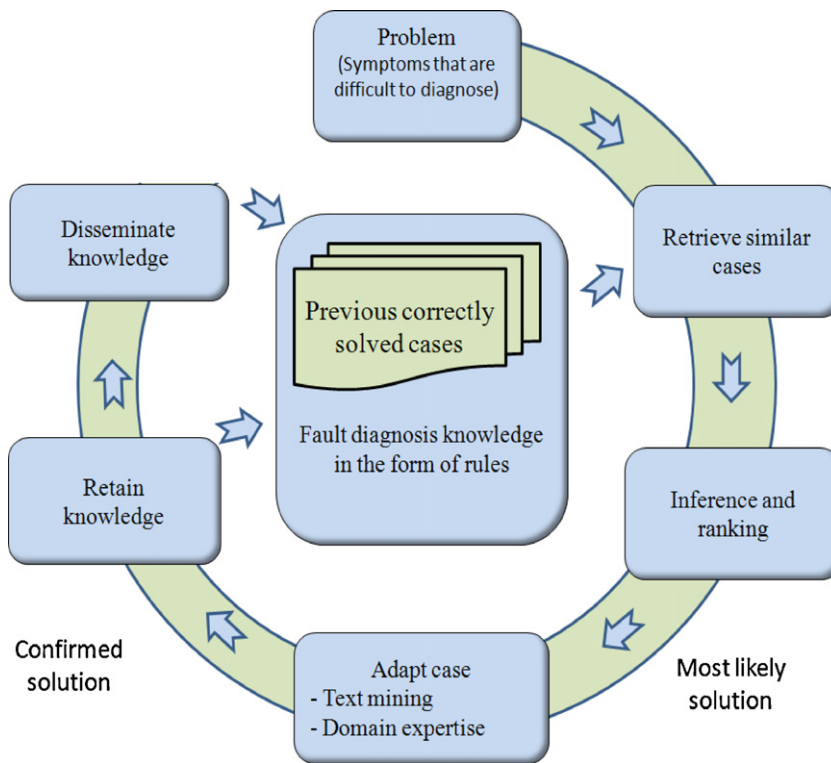


Fig. 2. The case-based reasoning component of the system.

knowledge in the form of rules which can be used to guide the reasoner to establish associations between the DTCs (i.e. symptom) at hand with the possible relevant labor codes. In contrast with the nearest neighbor algorithm which looks for similar case/cases from the case-base and apply its solution to the problem at hand, we aggregate past experience and knowledge from such similar cases (i.e. solutions) that can be applied to fix the DTCs at hand. This process of knowledge aggregation involves several preprocessing steps to remove the noise. In many instances, multiple DTCs are uploaded from various sources (such as OnStar,² Dealer diagnostics) over a period of time and also multiple repairs are performed on vehicle over a period of time. To deal with these, the temporal constraints in the form of rules are specified to restrict the scope of data such that the data residing only within a specific time window is considered for the knowledge aggregation. One such example of the rule is described below:

“the events observed for the same vehicle in Symptoms data are related to the events observed in Claims data if the lag between “Repair Date” and “DTC Read Date” is between $\pm x$ days, e.g. ± 5 days.”

Further, in many cases, technicians perform additional miscellaneous repairs/services (such as ‘courtesy transportation’, ‘Tire pressure check’) to maintain customer satisfaction. However, such repairs may not necessarily be directly related to any of the DTCs which technician is diagnosing and often introduce a noise. To alleviate this problem, we make use of the domain knowledge formalized in the form of rules. One such example of a rule states that, if a DTC is a powertrain related DTC then only consider the repairs which are related to engine and transmission in the process of aggregation. Such rules are crafted manually based on the domain knowledge. These rules are represented by deciphering the

codes corresponding to DTCs and LCs. For example, all power train related DTCs start with P [22] while the labor codes related to engine and transmission start with J and K respectively. Thus, if DTC starts with P then the labor codes starting with J and K are considered in the process of knowledge aggregation.

Having identified possible labor codes as solutions, a confidence value is assigned to them to determine to what extent a given labor code can be seen as appropriate to fix the symptom. The confidence value of 1 indicates that related labor code (repair) is used in all previous cases for a given symptom. The method that is used to compute the confidence is similar to the one described in Section 3.2 and Eq. (3). For example, the confidence of using labor code $L1$ for a string of DTCs say $DTC1$, $DTC2$, $DTC3$ is given as,

$$\text{Confidence}(DTC1, DTC2, DTC3 \Rightarrow L1) = \frac{N(L1, DTC1, DTC2, DTC3)}{N(DTC1, DTC2, DTC3)} \quad (5)$$

where,

$N(DTC1, DTC2, DTC3)$ = number of instances of diagnostic trouble codes ($DTC1, DTC2, DTC3$) set on a correctly diagnosed vehicle, $N(L1, DTC1, DTC2, DTC3)$ = number of instances $L1$ used to fix $DTC1$, $DTC2$, $DTC3$ occurring together.

In the case where more than one solution can be used as repair action, it is necessary to rank these solutions based on the confidence factor. Rank 1 indicates the most likely repair for the given symptoms. The case retrieval and ranking will lead to one of the following scenarios:

Scenario 1. The repair with rank 1, that is, the most likely repair has very high confidence factor and statistically significant number of similar cases are present in the case-base. The statistical significance is decided by the end user based on number of cases and own judgment.

² OnStar is a service network that uses wireless communication to provide diagnostic and navigation services to a driver (www.onstar.com).

Scenario 2. In some cases the top n repairs suggested by the system have the same confidence factor and, therefore, it becomes difficult to determine the most likely solution (repair).
Scenario 3. The number of cases that are retrieved in the retrieval phase are statistically not significant at all.

In the case of scenario 1, the solution suggested by the system can be used as a valid solution with a very little adaptation. Scenario 2 requires further case adaptation and learning. In the present approach, the case adaption and learning phase is performed interactively by the end user and the text mining module facilitates this process. This process is described in detail in the next sub-section. In the case of scenario 3, that is, in the absence of sufficient cases in the case-base, solutions to DTCs are evolved collaboratively by an offline consultation between team of domain experts from multiple disciplines (such as brand quality, service and product engineering).

3.3.2. Case adaptation and learning

For the set of solutions which require further adaptation, we first retrieve the technician verbatim entries associated with the labor codes. These verbatim consist of natural language description of repairs that are performed by technicians to fix the specific types of symptoms that are observed on a vehicle.

As mentioned in the last section, in some cases the feasible set of solutions that are suggested by the system may have the same confidence value and it becomes rather difficult to rank the solutions (Scenario 2). All the technician verbatim associated with the labor codes with the same confidence value are first collected and they are analyzed to establish more evidence from the text records to break the tie. In other words, we gather additional evidence by analyzing additional facts, that is, non-DTC related symptoms recorded in technician verbatim. As an example, for DTC P0300 (i.e. 'engine misfire detected'), let us assume that repairs 'spark plug replacement' and 'drive belt replacement' have same confidence level. However, other non DTC related symptom 'pitch tapping noise' has been found predominantly in technician verbatim for repairs related to 'drive belt replacement'. Thus, for vehicles with DTC P0300 and 'pitch tapping noise' the likeliest repair is 'drive belt replacement'.

The text mining component consists of following three sub-modules – document annotation engine, semantic extractor, and term based clustering engine (please refer to Fig. 3). The D&P (Diagnosis and Prognosis) ontology forms the back bone of text mining. The D&P ontology and different sub-modules of text mining are described in the next subsections.

3.3.2.1. Diagnosis and prognosis ontology. The D&P ontology consists of domain-specific knowledge of parts, part location, actions, and symptoms. It is formalized in RDF(s) (Resource Description Framework) [23]. The D&P ontology is constructed by making use of different databases, e.g. symptoms and claims, other textual sources like service manuals, technician service bulletins, and also by consulting with SMEs (Subject Matter Experts). At a coarse grained level, the D&P ontology is a structure of the form: $D\&P_{\text{onto}} := (C, C_{\text{subclass}}, \mathfrak{R}_{C_i}, I)$. The class, C , in the D&P ontology represents the top level concepts, such as Part, Action, PartLocation, Symptom, and LaborCode. The instance, I , formalizes the real world objects of domain specific concepts, for example, Door is an instance of the concept Part, etc. In many cases, the parts are written by using inconsistent vocabulary. For example, the part called 'Electronic Brake Control Module' can be written in the field data as either 'brake control module' or 'EBCM', and it is crucial to handle such terminological variations in the data. To handle such cases, each concept in the D&P ontology has a namespace called '<baseWord> Electronic Brake Control Module</baseWord>' and corresponding synonyms associated with it. The synonyms are also useful to handle the cases in which symptoms are recorded by using phrases, for example, 'cooling fan inoperative', 'cooling fan always on', 'cooling fan failing to shut off', and so on. The base words are used to disambiguate between inconsistent terminological references that are observed in the field data. The class PartLocation allows us to specify the precise location of each part in vehicle architecture, for example 'front left tire'. The class Action on the other hand represents different types of repair actions that can be performed to fix the symptoms. The class Symptom is used to formalize the clues that can be used to identify the nature of faults. This class is further specialized into two subclasses DTCRelatedSymptom and nonDTCRelatedSymptom. The more specific representation of the top level concepts is organized in terms of the class-subclass hierarchy, C_{subclass} . For instance, the parts such as 'engine control module', 'rear left door' provide more specific representation of the top level parts such as 'control module' and 'door', which appear in vehicle architecture. The top level class can have one or more subclasses associated with it. Finally, the binary relationships of the form $\mathfrak{R}_{C_i \rightarrow C_j}$ are defined in our ontology to formalize how two or more classes are associated with each other. Some of the main relationships in the D&P ontology involve – PartHasALocation (Part PartLocation), ActionPerformedOnPart (Action Part), SymptomAssociatedWithPart (Symptom Part), ActionRectifiesSymptom (Action Symptom), and ActionHasLaborCode (Action LaborCode). The existing

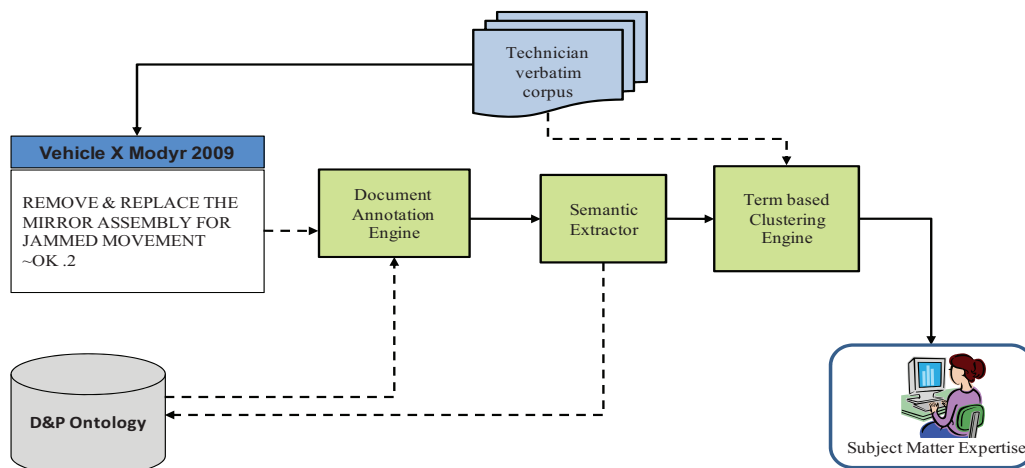


Fig. 3. The overall process.

version of the D&P ontology consists of 2508 parts, 78 actions, and 432 symptoms.

3.3.2.2. Document annotation engine. The *document annotation engine* correctly identifies parts, actions, and DTCs as well as non-DTC related symptoms that are recorded in the technician verbatim. To annotate the textual information, the document processing engine subscribes to the D&P ontology. For annotation, each technician verbatim in the corpus is first split into sentences and then tokenized. The stop words, like 'a', 'an', 'the', 'who', that are considered as non-descriptive are removed from the tokens to reduce the noise in the text. The filtered tokens are then stemmed using Porter stemming algorithm [24]. In theory, the lemmatization can be used to reduce the inflated form of the words that appear in a technician verbatim by taking into account the context. However, in many cases the field data is incomplete in nature, i.e. only the part and repair action is recorded but no information is provided about the symptoms, or written briefly. To this end, it becomes difficult to realize the context in which specific information is recorded. In future version of the text mining tool our aim is to make use of the missing data techniques (such as global imputation based on missing attributes) to transform incomplete data into complete technician verbatim, and then make use of lemmatization. The processed tokens are considered as a bag-of-words which are then matched with the concepts in the ontology. Initially, the algorithm performs full-string matching to see whether a token, say 'door', can be matched directly with the corresponding concept from the ontology. In cases the tokens cannot be matched directly because they are recorded by using abbreviations, say 'TPS'. To handle such cases, the abbreviation term, 'Term_i' to be disambiguated is identified from the text document and the base-words, that is 'Base-Word_j' and 'Base-Word_k' associated with 'Term_i' (for example 'tank pressure sensor' and 'tire pressure sensor') from the D&P ontology are selected. For each base-word the algorithm first identified the text documents in which it is recorded and then the word window of four terms is set on the either side of a selected base-word and the pairs of the form – Base-Word_j ∈ ({Base-Word_j Symptom_j}, {Base-Word_j Action_k}) and Base-Word_k ∈ ({Base-Word_k Symptom_j}, {Base-Word_k Action_k}) are constructed by taking into account the symptoms and the actions that appear in the word window of Base-Word_j and Base-Word_k. The conditional probabilities $P(\text{Base-Word}_j|\text{Symptom}_j)$, $P(\text{Base-Word}_j|\text{Action}_k)$ and $P(\text{Base-Word}_k|\text{Symptom}_j)$, $P(\text{Base-Word}_k|\text{Action}_k)$ are calculated to realize the probability of 'Base-Word_j' and 'Base-Word_k' appearing with Symptom_j and Action_k in a corpus. The set of pairs with highest conditional probability are treated as the valid pairs and the base-word, which is a member of the valid pair is established as the correct meaning of an abbreviation in the given context.

3.3.2.3. Semantic extraction engine. The *semantic extraction engine* takes as an input each annotated verbatim and extracts the annotated text from it. The semantic extraction engine provides a flexible way to extract semantics from the annotated technician verbatim in terms of different combinations like <part action>, <part symptom>, and a triple <part action symptom>. This makes sure that no information is overlooked that can be used by the clustering engine while generating different clusters.

3.3.2.4. Term based clustering engine. The *term based clustering engine* implements three novel hierarchical clustering algorithms to construct the clusters of technician verbatim such that it enables the end users to derive more evidence about the non-DTC related symptoms that are recorded in the technician verbatim. In other words, there are hundreds of thousands of technician verbatim data available in the field and for the end users it is crucial to

identify the technician verbatim cases that are associated with frequently failing parts along with symptoms that are associated with the parts and finally the repair actions that are used to fix the symptoms. To this end, three clustering algorithms are deployed – Part-based clustering, Part-Symptom based clustering, and Part-Symptom-Action based clustering. The Part-based clustering algorithm clusters the technician verbatim that are associated with each part, which helps the end users to identify the number of cases that are associated with each part failure in the field. Similarly, the Part-Symptom clustering algorithm clusters the technician verbatim which consists a record of a specific part along with symptoms that are associated with it, say 'battery-dead', 'battery-leak', which helps the end users to quickly identify frequent types of symptoms that are associated with each part failure. Finally, the Part-Symptom-Action clustering algorithm clusters the technician verbatim, which consist a record of a specific part along with symptoms that are associated with each part, and the repair actions that are used to fix the symptoms, say 'battery-dead-recharge' or 'bcm-internal_failure-replace'. This helps the end users to identify the frequently used repair actions from the data when a specific type of symptom is observed with a part.

The Part-based clustering algorithm takes as the input parts, part_i extracted by the *semantic extractor* along with the corpus of technician verbatim. Each technician verbatim is then treated as a bag-of-concepts and the frequency of each concept is calculated to determine the criticality of each part in a given corpus. Each technician verbatim is represented as a vector, $tv_f = \{tv_1, tv_2, \dots, tv_m\}$, where tv_m is the frequency of extracted semantic in a technician verbatim [25,26]. The agglomerative process of clustering is initialized by assigning a minimum frequency threshold of two to each part_i to determine the validity. The minimum frequency threshold in our case is set to two because in many cases three different types of verbatim namely *customer verbatim*, *correction verbatim* and *causal verbatim* are collected. The *customer verbatim* provides a description of the problem that is faced by a customer, for example 'reduced engine power'. The *correction verbatim* provides a description of the repair actions that are performed to fix the problem, for example 'Powertrain control module is replaced'. The *causal verbatim* provides a description of the root cause associated with the fault, for example, 'there was an internal short to Powertrain control module'. If part_i occurs only once in a single verbatim with no symptoms or actions co-occurring with part_i, then it provides limited contextual information to consider part_i as the valid field failure. Our main aim is to identify the part failures that are associated with the symptoms and the repair actions used to fix the faults, and, therefore, at any stage our algorithm relies on collecting as much contextual knowledge that is related with part_i in the form of symptoms or actions. Having identified the minimum frequency threshold, the two most similar clusters i.e. $C_{\{part_i\}} \cup C_{\{part_j\}} = C_{\{part_i, part_j\}}$ are merged and the process is iterated till there is one large cluster. For merging two clusters, $C_{\{part_i\}}$ and $C_{\{part_j\}}$ the distance between them is calculated by the average linkage.

The Part-Symptom clustering algorithm uses the clusters constructed by the Part-based algorithm and the pair {Part Symptom}_{sem-extract} that are extracted by the *semantic extractor* as the cluster label and these clusters are subdivided in such a way that each sub-cluster consists the cases of technician verbatim with a record of a specific part and the symptoms that are associated with it. Here, in each sentence of a technician verbatim first the focal term, that is, Part_i, is identified and the *word window* of four terms is set on the either side of Part_i to construct the pairs {Part_i Symptom_j} by taking into account the symptoms that are appearing in the word window (for example {battery dead}, {battery leakage}, and so on). In our approach, we take the

OLH Analysis				
LC Desc	No. Of Occurrences	No. Of Occurrences OLH>0 (N)	Total OLH in Hours (T)	Overall Contribution to OLH in Hours (T/T)
POWERTRAIN CONTROL MODULE REPLACEMENT	593	183	310.3	2.37
FRONT BRAKE ROTOR REFINISHING	5402	244	280.4	2.14
EMISSIONS SYSTEM WIRING AND/OR CONNECTOR REPAIR OR REPLACE	576	239	261.3	2
GOODYEAR/DUNLOP TIRE REPLACEMENT	5655	214	241.4	1.84
TRANSMISSION WIRING AND/OR CONNECTOR REPAIR OR REPLACEMENT	334	212	198.6	1.52
OCCUPANT SAFETY WIRING AND/OR CONNECTOR REPAIR OR REPLACEMENT	524	180	171.6	1.31

Fig. 4. Anomalies related to high time to repair.

viewpoint that the symptoms that are appearing with a part in the word window are more likely to be associated with a part in comparison with any other symptom that may be written in a document further away from a part. Each newly constructed pair $\{Part_i, Symptom_j\}$ is then matched with the label pair $\{Part, Symptom\}_{sem-extract}$ to make sure that it is a valid pair. Alternatively, if $\{Part_i, Symptom_j\}$ is not a member of the label pair $\{Part, Symptom\}_{sem-extract}$ then instead of directly using $\{Part_i, Symptom_j\}$ to construct the clusters its frequency is calculated to see whether it occurrence is greater than the *minimum frequency threshold* in the corpus. In the Part-Symptom-Action clustering algorithm, the clusters constructed by the Part-Symptom clustering algorithm along with the triples $\{Part, Symptom, Action\}_{sem-extract}$ are taken as the input. The new clusters are constructed in such a way that each cluster consist the cases of technician verbatim with the record of a specific part along with the symptoms that are associated with it and the repair actions used to fix the symptoms. Similar to the Part-Symptom clustering algorithm, in this algorithm, first the focal term, that is, $Symptom_j$ in each sentence of a technician verbatim is identified and the *word window* of four terms is set on the either side of $Symptom_j$ to identify the actions, $Action_k$, that are appearing in the *word window* to construct the triple of the form $\{Part_i, Symptom_j, Action_k\}$, for example, $\{Battery\ Dead\ Replace\}$, ..., $\{Battery\ Leakage\ Repair\}$. Each action appearing in a word window is deemed to be the one that is used to fix $Symptom_j$ in the field. Each newly constructed triple is then checked to see whether newly constructed triple matches with $\{Part, Symptom, Action\}_{sem-extract}$ and then its frequency is calculated to see whether it is greater than the *minimum frequency threshold*.

These clusters are presented to the end users to get their recommendations about whether the useful non-DTC related symptoms can be used in conjunction with DTCs to improve the confidence for a repair action suggested by system.

3.3.3. Knowledge retention and disseminations

The likely solutions/recommendations suggested by the system are validated by a domain expert and then retained in the knowledge base. Thus, as new knowledge gets added to the system, it progressively becomes richer and thereby minimizes the burden on end-user in the process of adaptation. Such a type of knowledge is considered as the best-practice knowledge in reducing unnecessary time and cost that may be spent by technicians to fix the anomalies. This knowledge is further disseminated to dealers to fix the similar symptoms for the future cases. Naturally, this helps to improve the performance of technicians such that these anomalies can be fixed accurately in a cost-effective way.

Thus, the case-base also gets richer as more number of correctly solved cases are continuously added and the system progressively becomes more and more intelligent with time.

4. Results and discussion

The overall approach has been implemented from scratch in a prototype system as a web-based distributed client-server architecture using the Java 2 Platform, Enterprise Edition (J2EE). The J2EE platform simplifies the development of enterprise applications using modular components like Enterprise JavaBeans (EJB), Java Servlets and Java Server Pages (JSP). The database has been implemented using Oracle database management system. The users from different groups (such as brand quality, product engineers and service) can access the database and perform the analysis. In actual practice, one brand quality manager is responsible to handle the service related issues and to maintain the knowledge base for each vehicle model. This helps to avoid inconsistencies in the process. The prototype version of the system is tested with input from the domain experts on the real-life data. Here, we first explain the working of the system with a case study.

4.1. Case study

We illustrate working of the system with a case study for reducing the time to repair. This case study is performed for a mid-sized sedan built and sold in North America. The top 5 repairs related to the high repair time identified by the system are shown in Fig. 4. An extra repair time (cf. Section 3.1.1) assigned to each labor code is described as Other Labor Hour (OLH). As shown in the figure, repair 'Power train control module (PCM) Replacement' is of the highest severity in terms of contribution to total OLH. The 'PCM replacement' operation has been performed in the field 593 times. Out of these 593 cases, in 183 cases an extra repair time has been assigned by the technicians and total extra repair time corresponding to these 183 cases is 310.3 h. The total extra repair time

Table 3
Association mining results for 'PCM Replacement'.

DTCs	Count	Percentage of cases	Total OLH (Hours)
P0315,P0633,P0513	93	51	192.7
P0315,P0633	34	19	73.4
P0601	7	4	11.5
P2135	4	2	5.5
P0513	2	1	1.2

	S.No	LC Desc	Count	Confidence
☐	1	POWERTRAIN CONTROL MODULE REPLACEMENT	213	1

Fig. 5. Reasoning for DTCs 'P0315, P0513, P0633'.

corresponding to all labor codes was 13080 h. Hence, contribution of labor code 'PCM replacement' is 2.37% (i.e. $310.3 \times 100/13080$).

The association rule mining technique has been used to further investigate into the root cause of high repair time cases. As mentioned before, this has been achieved by finding the associations between the Symptoms (DTCs) and the Claims related to the anomalous cases. Results of the association rule mining for the anomalous cases related to labor operation 'PCM Replacement' are shown in Table 3. In 51% of the cases, an extra repair time has been taken by the technicians when the DTCs P0315, P0633 and P0513 occur together.

The learning for the DTCs 'P0315, P0633, P0513' has been performed for the non anomalous cases (Fig. 5). As it can be realized from the figure, when DTCs 'P0315, P0633, P0513' occur together PCM has been replaced in all the cases. Under such circumstances it can be concluded that, the root cause for the DTCs 'P0315, P0633, P0513' is PCM which needs to be diagnosed and replaced. From anomalous cases, it was also realized that technicians (especially, inexperienced technicians) tend to diagnose these DTCs separately, which in turn increases the diagnosis time. The discovered knowledge i.e. "when the DTCs 'P0315, P0633, P0513' occur together first diagnose PCM" was communicated to the technicians to reduce the diagnostic time and cost.

Thus, the anomaly detection and the root cause investigation components provide analytics capabilities to identify the issues in the field and perform its root cause investigation, while reasoning provides a decision support to reduce the anomalies. The developed system is evaluated to see whether the solutions provided by the system (reasoning component) indeed concur with those provided by the domain expert. We also investigated into the effect of addition of domain specific knowledge (cf. Section 3.3.1) in improving the performance of the reasoner. In addition, the entire system has been evaluated for the scalability in real life industrial set up.

4.2. Performance evaluation

In this section, we first describe the evaluation strategy for the case-based-reasoner followed by related results. Then, we present the results for the scalability evaluation.

Table 4

Comparative analysis of the solutions retrieved by reasoner.

DTC symptoms	Number of likely solutions suggested by domain expert (n)	Case retrieval and reasoning with domain knowledge				Case retrieval and reasoning without domain knowledge			
		Number of retrieved solutions	Rank of most likely solutions	Number of relevant solutions	Precision/recall	Number of retrieved solutions	Rank of most likely solution	Number of relevant solutions	Precision/recall
P0300	5	5	1	5	100%	5	1	3	60%
P0455	3	3	2	3	100%	3	2	2	67%
P0463	3	3	1	3	100%	3	1	2	67%
P0315, P0513, P0633	1	1	1	1	100%	1	1	1	100%
P0172	7	7	1	7	100%	7	1	5	71%

4.2.1. CBR evaluation strategy

From the anomaly detection and the root cause investigation modules, it was realized that certain DTCs are difficult to diagnose in the field leading to high time to repair or repeat visits. One such set of DTCs are listed in Table 4. We presented these DTCs to the domain experts and asked them to propose n likely solutions for these DTCs and rank them based on their likelihood using their experience. We then used these ranked solutions as a baseline to evaluate the performance of the CBR component of system.

The same sets of DTCs are then used by the case retrieval and learning component to retrieve the past cases from the case-base to find the likely solutions and rank them based on their confidence factor. We then compared and contrasted the likely solutions proposed by our system with the solutions suggested by the domain expert.

The retrieval performance of the system is evaluated based on measures precision and recall. The precision is the fraction of retrieved solutions that are relevant to the symptoms, whereas, the recall is the fraction of the relevant solutions that are successfully retrieved.

$$\text{Precision} = \frac{|\text{relevant solutions extracted}|}{|\text{set of extracted solutions}|} \quad (7)$$

$$\text{Recall} = \frac{|\text{relevant solutions extracted}|}{|\text{total relevant solutions}|} \quad (8)$$

In our experiments, we restricted solution retrieval to top n solutions (i.e. number of relevant solutions suggested by the domain expert). This leads to the same value for the precision and the recall. In these experiments, we also evaluated whether the most likely solution suggested by the system (i.e. solution with highest confidence) is in compliance with the most likely solution suggested by the domain expert. Several such experiments were performed to test the system.

As mentioned earlier, to improve the performance of our system, we incorporated the domain knowledge formalized in the form of rules (cf. Section 3.3.1). We also investigated into the performance improvement of the reasoner by addition of the domain specific knowledge. For this, we repeated the above experiments to find the likely solutions for the same set of DTCs without rule-based domain knowledge. The solutions were once again compared with the solutions suggested by the domain expert to determine the precision (or recall). Finally, the precision values for solution retrieval without the domain knowledge are compared with the precision values for solution retrieval with domain knowledge to identify the improvement.

4.2.2. CBR evaluation results

Several experiments are performed to study the effectiveness of solution retrieval and reasoning. The results of these experiments are summarized in Table 4. The performance of the system in terms

	S.No	LC Desc	Count	Confidence
☐	1	MASS AIRFLOW SENSOR WITH INTAKE AIR TEMPERATURE SENSOR REP	103	0.21
☐	2	DUCT, AIR FLOW SENSOR TO THROTTLE BODY - REPLACE	93	0.19
☐	3	09298 - CONTROL ASSEMBLY-RADIO AUDIO (FACE PLATE) RETURN T	80	0.17
☐	4	AIR CLEANER OUTLET RESONATOR REPLACEMENT	42	0.09
☐	5	09298 - FOR ESC USE ONLY - REPROGRAM ICS	32	0.07
☐	6	AIR CLEANER ASSEMBLY REPLACEMENT	27	0.06
☐	7	ENGINE CONTROL MODULE REPROGRAMMING WITH SPS	21	0.04

a

	S.No	LC Desc	Count	Confidence
☐	1	MASS AIRFLOW SENSOR WITH INTAKE AIR TEMPERATURE SENSOR REP	103	0.26
☐	2	DUCT, AIR FLOW SENSOR TO THROTTLE BODY - REPLACE	93	0.23
☐	3	AIR CLEANER OUTLET RESONATOR REPLACEMENT	42	0.11
☐	4	AIR CLEANER ASSEMBLY REPLACEMENT	27	0.07
☐	5	ENGINE CONTROL MODULE REPROGRAMMING WITH SPS	21	0.05
☐	6	FUEL INJECTOR REPLACEMENT	15	0.04
☐	7	POWERTRAIN CONTROL MODULE TRANSMISSION REPROGRAMMING WITH	12	0.03

b

Fig. 6. Reasoning for DTC P0172 (a) without domain knowledge and (b) with shallow domain knowledge.

of solution retrieval (precision and recall) is as good as the domain expert. In all cases, solutions suggested by system are matching with the solutions suggested by the domain expert. It was observed that in most of the cases, the most likely solutions suggested by the system matched with the most likely solutions suggested by the domain expert. However, in case of the DTC P0455, the most likely solution from the field data was not in compliance with the solutions proposed by the domain expert. Thus, the system also enabled to discover new knowledge (which is again validated by the domain expert) from the data. In such scenario, the newly validated knowledge can be transferred to dealers to diagnose the problem in the sequence that was suggested by the system to reduce the diagnosis and repair time. In real practice, this dissemination of knowledge happens through service manual or technical service bulletins (TSBs). These service manuals or TSBs explain step by step instructions about disassembling a particular system or subsystem, diagnosis procedure and repair action to rectify the symptoms. The new knowledge discovered from the case-based-reasoning is used to improve the service manual sequence (to improve the mean time to repair) and content (to reduce repeat visits). New failure modes and actions discovered from the field are added to the service manual to fix the problem 'right first time'.

The performance improvement by incorporating the domain knowledge is illustrated with an example of DTC P0172. In the absence of the domain knowledge (Fig. 6a), the most likely repair suggested by the system for DTC P0172 is 'mass airflow sensor with intake air temperature sensor replacement.' This repair was in compliance with the most likely repair suggested by the domain expert. However, two repairs suggested by the system (i.e. repairs related to 'control assembly radio' and 'reprogram ICS') were not in compliance with those suggested by the domain expert. Further, some of the repairs suggested by the domain expert i.e. repair related 'fuel injector' and 'transmission reprogramming' were not present in the solutions suggested by the system. The incorpo-

ration of domain knowledge helped to remove noise and also all the repairs that were suggested by the system were in compliance with those suggested by the domain expert (Fig. 6b).

The results for improvement in the solutions retrieval for various cases are shown in Table 4. In the worst case (i.e. for DTC P0300), about 40% of the top n solutions retrieved by the system did not appear in the list of the likely solutions proposed by the domain expert. The overall improvement of 0–40% was achieved in the systems' performance by using knowledge guided (domain knowledge in the form of rules) retrieval.

In some cases, we realized that the top solutions, i.e. labor codes say $L1$ and $L2$ retrieved by the system showed the same confidence value, and as a result it was difficult to rank the solutions. To deal with such situation, the text mining component of the system was used to get more fine-grained insight into the problem. Here, first the technician verbatim records associated with $L1$ and $L2$ were retrieved and correctly annotated to highlight the non-DTC related symptom (e.g. leak or noise) recorded by the technicians. Such additional information was used for performing further analysis (Fig. 7). The document annotation engine and semantic extractor component of the system successfully managed to retrieve all the non-DTC related symptoms correctly without missing any one of them. The term based clustering engine clustered the set of technician verbatim in which the non-DTC related symptoms extracted by the system were recorded. Each cluster of technician verbatim generated for non-DTC related symptoms were analyzed interactively by the end users. These non-DTC related symptoms that appeared in the highest number of technician verbatim entries in conjunction with DTCs were used to assign priorities for labor codes $L1$ and $L2$. In practice, it was observed that there were multiple linguistic variations to describe the same symptom in technician verbatim entries. For example, the symptom 'noise' was also recorded as 'rattle' or 'clink'. The D&P ontology consisted synonyms of symptoms, helped to disambiguate between different linguistic variations and improved the clustering performance of the system.

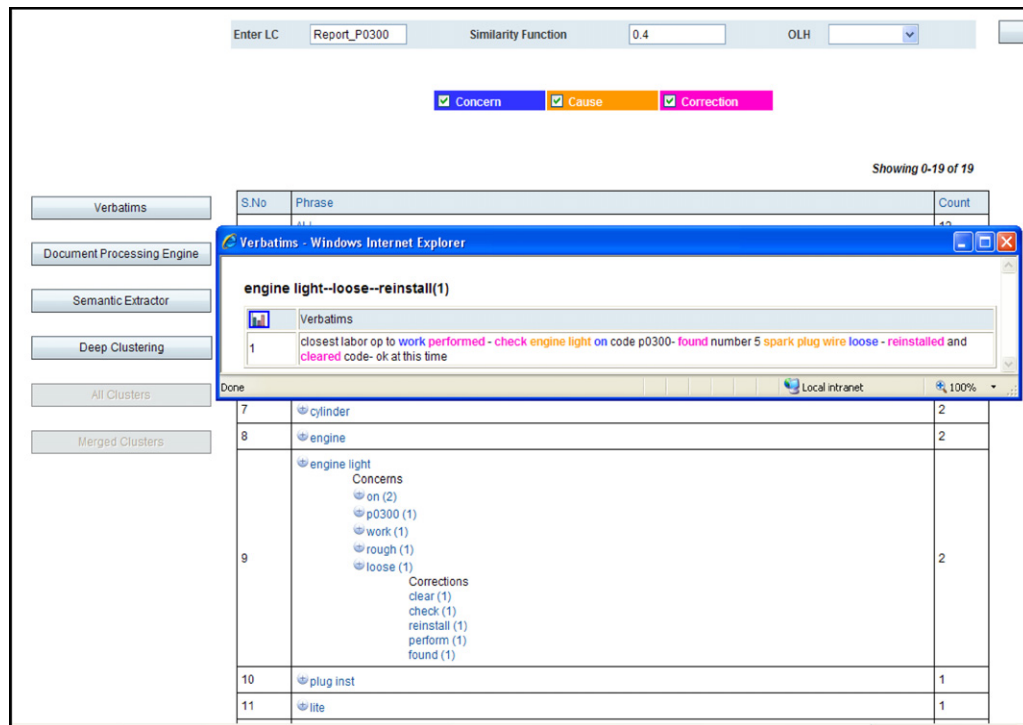


Fig. 7. A sub-cluster of technician verbatim with part symptom pair.

Table 5
Scalability performance of the system.

Claim selection period	Number of records in Claims database	Time taken to identify anomalous cases related to high repair time (s)	Number of cases related to LC contributing highest to high repair time	Time taken to perform root cause investigation to identify DTCs related to LC (s)	Time taken to perform reasoning and learning (s)
One month	218,095	25	23	8	32
Three months	715,132	38	148	17	74
One year	3,425,293	69	362	38	133

4.2.3. Scalability

The scalability is one of the most important criteria to determine the usefulness of the system in a real environment. We performed a set of experiments with the data sets of different sizes for different vehicle makes and models to see amount of time taken by the system to – (a) identify the anomalous cases and the labor codes contributing to high repair time, (b) perform the root cause investigation of the labor code contributing maximum to high repair time and identify the culprit symptoms (DTCs), and (c) perform learning and reasoning for these DTCs. All the experiments were performed on OS Type: Microsoft Windows XP Professional, Memory size: 3.5 GB, Processor: Intel[R] Core[TM] 2 Duo CPU T9300 @ 2.50 GHz 2.49 GHz. Table 5 gives details of this evaluation. Before implementing our system, it was not possible for the end users to analyze the field data efficiently and in many cases the crucial anomalies were overlooked. It was observed that, the time required to perform such analysis was reduced from three to four weeks to few minutes by using our system.

5. Conclusions

In this paper, we developed a novel integrated approach that uses the association rule mining, CBR and text mining for continuous improvement in the service and repair. The developed methodology has been implemented in a prototype framework to identify the anomalies in the field, perform its root cause analysis

and to identify the training needs of the service technicians. Moreover, it also facilitates the learning the best practices and the discovery of new knowledge to achieve an overall improvement in the service.

The existing version of the system was tested on the real word automotive data related to millions of vehicles. A significant improvement in term of time has been observed to detect the anomalies and learn from the best practices. It was realized that, with the developed integrated approach the time required to perform the analysis and learn from the field was reduced from few weeks to few minutes. More importantly, it also identified the anomalies and the best practices in the field which were overlooked because of the overwhelming data size. In future, our aim is to automate the process of case adaptation and learning to reduce the burden on end users. This involves, automatically extracting the domain knowledge from other sources such service manuals, service bulletins, engineering documents and compare the knowledge from these sources with the field data to find the anomalies as well as discover new knowledge. Further, the work on the association rule mining for repeat visits can be extended to multiple visits to identify the sequence of repairs that are involved in the anomalous cases. Nevertheless, the existing version successfully helped to identify the wastes that cause customer inconvenience, perform its root cause investigation and learn the best practices to achieve continuous improvement and the lean service in the automotive domain.

The developed approach can be applied to improve the diagnosis and repair in various domains (such as manufacturing, aerospace, medical) where data related to the symptoms and the remedial (repair) actions exist. In the manufacturing plants, the failures and the symptoms are recorded in 'Down Time Reporting (DTR)³' systems. The repairs performed by the plant maintenance technicians to fix the symptoms are recorded in maintenance management systems [27]. The developed approach can be applied to such manufacturing data to reduce the machine down time and to improve the overall operational efficiency of the plant. In the medical domain, electronic medical records [28] can be mined to investigate into the symptoms that are difficult to diagnose causing inconvenience to the patients (example, repeated visits) and learn the best practices to treat such symptoms. This will help to reduce the diagnostic cost and visits to hospital.

To the best of our knowledge, a seamless integration of the association rule mining, case-based-reasoning and text mining techniques for improving the service is the first of its kind, and we hope that it will spur further research in the manufacturing, aerospace and medical communities.

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³ <http://www.premier-system.com/files/SS%20Auto-IT%20Downtime%20Reporting%20DTR.pdf>.